National University of Computer and Emerging Sciences

Islamabad Campus

Probability & Statistics Sessional-1 Solution

Question # 1 **[13 Marks]**

A quality control analyst recorded the daily processing time (in minutes) required to complete a specific manufacturing task over ten randomly selected days. The observations are:

$$18,\ 22,\ 19,\ 24,\ 27,\ 31,\ 29,\ 34,\ 42,\ 95$$

(a) Obtain the five-number summary (maximum, minimum, median, Q_1, & Q_3) and construct a box-and-whisker plot. Based on the graphical structure, comment on the symmetry of the distribution. (7)

Sorted data: 18, 19, 22, 24, 27, 29, 31, 34, 42, 95

Minimum = 18 Maximum = 95

Median = (27 + 29)/2 = 28

Q1 = Median of first five = 22

Q3 = Median of last five = 34

Five-number summary: (18, 22, 28, 34, 95)

The distribution is positively skewed (right skewed) due to large value 95.

(b) Investigate whether any observation may be classified as an outlier. (3)

IQR = Q3 - Q1 = 34 - 22 = 12

Lower Bound = 22 - 1.5(12) = 4

Upper Bound = 34 + 1.5(12) = 52

Since 95¿52, therefore 95 is an outlier.

(c) Suppose the original mean processing time is $\bar{x} = 34.1$ minutes and the original variance is $s^2 = 530.1$. Management adjusts the processing time records by first adding 5 minutes to each observed time, and then multiplying the result by 2. Without recalculating from the raw data, explain how this change affects the original mean and variance. (3)

If Y = 2(X + 5) = 2X + 10

New Mean: = 2 + 10

New Variance: Var(Y) = 4 Var(X)

Adding constant shifts mean. Multiplying by 2 multiplies variance by 4.

Question # 2 **[10 Marks]**

A questionnaire on housing arrangements showed this information obtained from 25 respondents. Construct a frequency distribution for the data (H = house, A apartment, M = mobile home, C = condominium).

$$H,\ C,\ H,\ M,\ H,\ A,\ C,\ A,\ M,\ C,\ M,\ C,\ A,\ M,\ A,\ C,\ C,\ M,\ C,\ C,\ H,\ A,\ H,\ H,\ M$$

(a) Construct a bar graph for the data. (3)

(b) Construct a pie graph for the data. (4)

(c) In Sessional-1 Exam of Calculus it has been observed that marks follow a normal distribution. It is also noticed that $\mu = 18$ and $\sigma = 2$ then calculate the intervals for this class in which

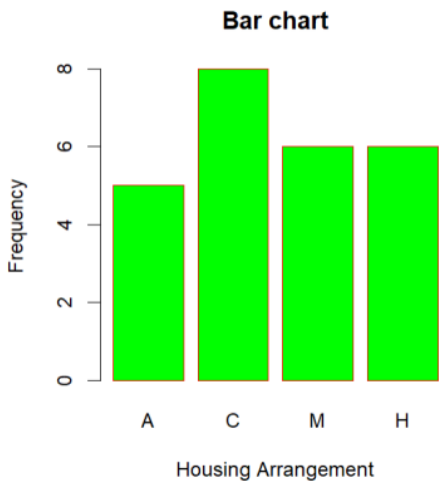
(i) 68% of data lies (1)

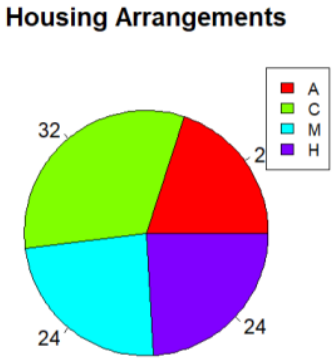
(ii) 95% of the data lies (1)

(iii) 99.7% of the data lies (1)

Solution:

Categories	Frequency	Relative Frequency
A	5	0.20
C	8	0.32
M	6	0.24
H	6	0.24





(C) (i) [16,20] **(ii) [14,22]** **(iii) [12,24]**

Question-3: **[21 Marks]**

(a) What is the difference between independent events and mutually exclusive events? Can two events A and B be both independent and mutually exclusive at the same time? (2)

Independent events:

$$P(A \cap B) = P(A)P(B)$$

Mutually exclusive:

$$P(A \cap B) = 0$$

They cannot be both unless one event has probability zero.

(b) Explain the difference between $P(A|B)$ and $P(B|A)$. Can they be equal? If so, under what conditions? (2)

$$P(A|B) = P(A \cap B)/P(B)$$

$$P(B|A) = P(A \cap B)/P(A)$$

They are equal if P(A) = P(B).

(c) If A_1, A_2 and A_3 are collectively exhaustive events(make up the entire sample space), what is the value of $P(A_1 \cup A_2 \cup A_3)$? (1)

Since A_1, A_2, and A_3 are collectively exhaustive events, their union equals the entire sample space S:

$$A_1 \cup A_2 \cup A_3 = S$$

We know that:

$$P(S) = 1$$

Therefore,

$$P(A_1 \cup A_2 \cup A_3) = 1.$$

(d) A Hackathon team is being formed to build a Smart Recommendation System. There are 5 Programmers, 4 Data Analysts, 3 UI Designers.

(i) In how many ways can a team of 5 members be selected from the 12 participants? (2)

$$\binom{12}{5} = 792$$

(ii) In how many ways can a team of 5 be formed such that it includes: 2 Programmers, 2 Data Analysts, 1 UI Designer? (2)

Number of Programmers = 5
Number of Data Analysts = 4
Number of UI Designers = 3
We must select: 2 Programmers, 2 Data Analysts, and 1 UI Designer.
Since selections are independent across categories, we use combinations:

$$\binom{5}{2} \times \binom{4}{2} \times \binom{3}{1}$$

Now compute each term:

$$\binom{5}{2} = 10, \quad \binom{4}{2} = 6, \quad \binom{3}{1} = 3$$

Therefore,

$$\binom{5}{2}\binom{4}{2}\binom{3}{1} = 10 \times 6 \times 3 = 180$$

(iii) Let A and B are events that team contains

$$A = \text{at least one UI Designer}, \quad B = \text{no Data Analyst}$$

Find the following probabilities: (5)

- $P(B)$
- $P(A)$
- $P(A^c)$
- $P(A \cap B)$
- $P(A \cup B)$

Total number of possible teams:

$$\binom{12}{5} = 792$$

1. Probability of B (No Data Analyst)

If no Data Analyst is selected, all 5 must come from 5 Programmers + 3 UI Designers = 8 people.

$$P(B) = \frac{\binom{8}{5}}{\binom{12}{5}} = \frac{56}{792} = \frac{7}{99}$$

2. Probability of A (At least one UI Designer)

First compute complement (no UI Designer selected).

If no UI Designer is selected, all 5 must come from 5 Programmers + 4 Data Analysts = 9 people.

$$P(A^c) = \frac{\binom{9}{5}}{\binom{12}{5}} = \frac{126}{792}$$

$$P(A) = 1 - P(A^c) = 1 - \frac{126}{792} = \frac{666}{792} = \frac{111}{132}$$

3. Probability of A^c

$$P(A^c) = \frac{126}{792}$$

4. Probability of $A \cap B$

Event B means no Data Analyst. So we choose 5 from 8 (5P + 3UI). To satisfy A (at least one UI), subtract the case of no UI.

Total ways with no Data Analyst:

$$\binom{8}{5} = 56$$

Ways with no Data Analyst AND no UI: (all 5 from 5 Programmers)

$$\binom{5}{5} = 1$$

Therefore,

$$P(A \cap B) = \frac{\binom{8}{5} - \binom{5}{5}}{\binom{12}{5}} = \frac{56 - 1}{792} = \frac{55}{792}$$

5. Probability of $A \cup B$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{666}{792} + \frac{56}{792} - \frac{55}{792}$$

$$= \frac{667}{792}$$

$$\boxed{P(B) = \frac{56}{792},\ P(A) = \frac{666}{792},\ P(A^c) = \frac{126}{792},\ P(A \cap B) = \frac{55}{792},\ P(A \cup B) = \frac{667}{792}}$$

(e) A cloud computing company monitors two core databases, Database A and Database B, for daily synchronization errors. Historical data reveals the following:

- The probability that Database A experiences an error on a given day is 0.40.
- The probability that Database B experiences an error is 0.30.
- The probability that at least one of the databases experiences an error is 0.60.

(i) Calculate the probability that both databases experience an error on the same day. (3)

$P(A \cap B) = 0.4 + 0.3 - 0.6 = 0.1$
Check independence: 0.4 0.3 = 0.12 0.1
Therefore NOT independent.
Conditional probability: P(B—A) = 0.1 / 0.4 = 0.25

(ii) Are the errors in Database A and Database B independent events? Prove your answer mathematically. (2)

Given:

$$P(A) = 0.40, \quad P(B) = 0.30, \quad P(A \cup B) = 0.60$$

First compute $P(A \cap B)$ using the addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.60 = 0.40 + 0.30 - P(A \cap B)$$

$$0.60 = 0.70 - P(A \cap B)$$

$$P(A \cap B) = 0.70 - 0.60 = 0.10$$

Now check independence.
Events A and B are independent if and only if:

$$P(A \cap B) = P(A)P(B)$$

Compute:

$$P(A)P(B) = (0.40)(0.30) = 0.12$$

Since:

$$P(A \cap B) = 0.10 \neq 0.12 = P(A)P(B)$$

Therefore,

$$\boxed{\text{The events are NOT independent.}}$$

(iii) On a day where it is known that Database A has experienced an error, what is the probability that Database B also experiences an error? (2)

We are given:

$$P(A) = 0.40, \quad P(B) = 0.30, \quad P(A \cup B) = 0.60$$

First compute $P(A \cap B)$:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= 0.40 + 0.30 - 0.60 = 0.10$$

Now compute the conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.10}{0.40} = 0.25$$

$$\boxed{P(B \mid A) = 0.25}$$

Thus, given that Database A has an error, the probability that Database B also has an error is 0.25 (or 25%).

Question-4 **[13 Marks]**

(a) If $P(A|B) > P(A)$, what does this imply about the relationship between A and B? (2)

Given:

$$P(A \mid B) > P(A)$$

By definition of conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Substituting:

$$\frac{P(A \cap B)}{P(B)} > P(A)$$

Multiply both sides by $P(B) > 0$:

$$P(A \cap B) > P(A)P(B)$$

Now recall:

If

$$P(A \cap B) = P(A)P(B)$$

then A and B are independent.

Since here:

$$P(A \cap B) > P(A)P(B),$$

this implies that the occurrence of B increases the probability of A.

Conclusion:

Events A and B are **positively associated** (positively dependent).

That is, the occurrence of B makes A more likely to occur.

Equivalently,

$$P(B \mid A) > P(B)$$

Thus, A and B have a positive relationship.

(b) To properly use the Law of Total Probability, you must divide your scenario into different underlying events or groups (often called partitions). What two strict mathematical conditions must these partitions satisfy for the law to work? (2)

1. Mutually exclusive 2. Collectively exhaustive

(c) In the context of Bayes' Rule, what is the conceptual difference between a "prior" probability and a "posterior" probability? (2)

Prior: initial belief before evidence.

Posterior: updated probability after evidence.

(d) A spam detection system classifies incoming emails as either Spam or Not Spam. It is observed that 30% of all emails received are Spam and 70% of all emails received are Not Spam. The system generates a Warning Signal whenever it suspects that an email is spam. If an email is actually Spam, the system correctly raises a warning 90% of the time. If an email is Not Spam, the system still raises a warning 20% of the time.

(i) Compute the overall probability that the system raises a warning. (4)

P(W) = 0.9(0.3) + 0.2(0.7) = 0.27 + 0.14 = 0.41

(ii) If the system raises a warning, compute the probability that the email is actually spam. (3)

$P(S|W) = 0.27/0.41 = 0.6585$

Question-5 [8 Marks]

(a) Why is the Naive Bayes algorithm called "naive"? What strong assumption does it make about the data? (1)

Called naive because it assumes conditional independence of features.

(b) In Naive Bayes, how is

$$P(X_1, X_2 \mid Y)$$

simplified? (1)

$P(X_1, X_2|Y) = P(X_1|Y)P(X_2|Y)$

(c) A hospital uses an AI system to detect a certain disease D. The AI checks two symptoms:

F_1 : High fever $\quad$ F_2 : Skin rash

The symptoms are assumed to be conditionally independent given the disease status. The following conditional probabilities are known:

$$P(F_1 \mid D) = 0.9, \quad P(F_1 \mid D^c) = 0.2$$
$$P(F_2 \mid D) = 0.8, \quad P(F_2 \mid D^c) = 0.1$$

The prior probability of the disease is $P(D) = 0.05$. If a patient has both high fever and skin rash, what is the probability that the patient actually has the disease? (6)

We want to compute:

$$P(D \mid F_1 \cap F_2)$$

Using Bayes Theorem:

$$P(D \mid F_1 \cap F_2) = \frac{P(F_1 \cap F_2 \mid D)P(D)}{P(F_1 \cap F_2)}$$

Step 1: Use conditional independence

Since symptoms are conditionally independent given disease status,

$$P(F_1 \cap F_2 \mid D) = P(F_1 \mid D)P(F_2 \mid D) = (0.9)(0.8) = 0.72$$

Similarly,

$$P(F_1 \cap F_2 \mid D^c) = P(F_1 \mid D^c)P(F_2 \mid D^c) = (0.2)(0.1) = 0.02$$

Step 2: Compute total probability (Law of Total Probability)

$$P(F_1 \cap F_2) = P(F_1 \cap F_2 \mid D)P(D) + P(F_1 \cap F_2 \mid D^c)P(D^c)$$

Since:

$$P(D) = 0.05, \quad P(D^c) = 1 - 0.05 = 0.95$$

$$P(F_1 \cap F_2) = (0.72)(0.05) + (0.02)(0.95)$$

$$= 0.036 + 0.019 = 0.055$$

Step 3: Apply Bayes Formula

$$P(D \mid F_1 \cap F_2) = \frac{(0.72)(0.05)}{0.055} = \frac{0.036}{0.055} = 0.6545$$

Final Answer:

$$\boxed{P(D \mid F_1 \cap F_2) \approx 0.655}$$

Interpretation:

Even though the prior probability of the disease was only 5%, after observing both symptoms, the probability increases to approximately 65.5%.